

Course Code: B23C14101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AD-HOC AND SENSOR NETWORKS					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Mobile Ad Hoc Network (MANET) and identify its key characteristics.	1	2	2
	b).	Summarize the challenges faced in MANET routing.	1	3	2
	c).	Identify the basic principle of the DSDV routing protocol .	2	3	2
	d).	Explain the concept of broadcast storm problem in MANETs.	2	2	2
	e).	Define geocasting and state its purpose in MANETs.	3	2	2
	f).	Explain why TCP performance degrades in ad hoc networks.	3	2	2
	g).	List the major components of a Wireless Sensor Network architecture.	4	3	2
	h).	Explain the role of the MAC layer in WSN .	4	2	2
	i).	Identify the functions of the transport layer in WSN .	5	3	2
	j).	State the importance of sensor networks in mobile robotic systems .	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Illustrate the characteristics, applications, and challenges of Mobile Ad Hoc Networks.	1	2	5
	b).	Compare and analyze proactive, reactive, and hybrid routing protocols used in MANETs.	1	3	5
		OR			
3.	a).	Explain the working principle of DSR and AODV routing protocols.	1	2	5
	b).	Analyze the functioning of position-based routing algorithms including DREAM and LAR.	1	3	5
		UNIT-2			
4.	a).	Summarize broadcast storm problem and analyze various rebroadcasting schemes.	2	3	5
	b).	Examine the multipoint relaying (MPR) technique used in broadcast control.	2	3	5
		OR			
5.	a).	Analyze tree-based and mesh-based multicasting protocols in MANETs.	2	3	5

	b).	Demonstrate the working of ODMRP and CAMP multicast protocols.	2	2	5
		UNIT-3	3	2	5
6.	a).	Compare the concept of geocasting in MANETs and its applications.	3	3	5
	b).	Analyze the LBM geocasting mechanism.			
		OR			
7.	a).	List out the challenges of TCP over MANETs .	3	2	5
	b).	Evaluate different solutions proposed to improve TCP performance in ad hoc networks.	3	3	5
		UNIT-4			
8.	a).	Illustrate the architecture and classification of Wireless Sensor Networks.	4	2	5
	b).	Analyze the MAC layer and link layer issues in WSNs.	4	3	5
		OR			
9.	a).	Explain the physical layer design considerations in sensor networks.	4	2	5
	b).	Analyze routing protocols used in Wireless Sensor Networks.	4	3	5
		UNIT-5			
10.	a).	Outline the transport layer protocols used in WSNs.	5	2	5
	b).	Analyze application layer support mechanisms in sensor networks.	5	3	5
		OR			
11.	a).	Explain how WSNs adapt to the dynamic nature of sensor environments.	5	2	5
	b).	Evaluate the role of sensor networks in mobile robotic systems.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23HS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRODUCTION & OPERATIONS MANAGEMENT					
(Common to CIC & CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Production & Operations Management in the context of a modern enterprise.	1	1	2
	b).	State the difference between the 'Controllable' and 'Uncontrollable' variables influencing an operations system	1	1	2
	c).	State two distinct characteristics of a Job Shop Production System.	2	1	2
	d).	Define 'Floor Plan Mapping' in facility design.	2	1	2
	e).	Define Supply Chain Management	3	1	2
	f).	What is meant by lead time in inventory management?	3	1	2
	g).	What is Kaizen?	4	1	2
	h).	What is Statistical Process Control	4	1	2
	i).	What is PERT.	5	1	2
	j).	Define critical path.	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.		Explain the Variables influencing the Production & Operations Management.	1	3	10
		OR			
3.		Discuss the evolution of Operations Management with historical perspectives.	1	3	10
		UNIT-2			
4.		Discuss the factors influencing the Site selection Criteria	2	3	10
		OR			
5.		Distinguish between mass, batch and continuous production systems.	2	3	10
		UNIT-3			

6.	a).	Discuss the ABC analysis technique in inventory control	3	3	5																	
	b).	Explain the Just-In-Time (JIT) inventory system	3	3	5																	
		OR																				
7.		Discuss the qualitative and quantitative demand forecasting techniques	3	3	10																	
		UNIT-4																				
8.	a).	Describe the benchmarking process and list its advantages	4	3	5																	
	b).	Explain the Six Sigma approach to quality improvement.	4	3	5																	
		OR																				
9.	a).	Discuss the ISO series of quality standards	4	3	5																	
	b).	Describe the principles of Total Quality Management	4	3	5																	
		UNIT-5																				
10.	a).	Discuss the techniques of capacity planning.	5	3	5																	
	b).	Describe the strategic production planning process	5	3	5																	
		OR																				
11.		Draw the network and find critical path, the minimum time to complete project and total float of each activity.																				
		<table><tr><td>ACTIVITY</td><td>1-2</td><td>1-3</td><td>2-3</td><td>2-4</td><td>3-4</td><td>5-4</td><td>5-3</td></tr><tr><td>TIME</td><td>20</td><td>25</td><td>10</td><td>12</td><td>6</td><td>10</td><td></td></tr></table>	ACTIVITY	1-2	1-3	2-3	2-4	3-4	5-4	5-3	TIME	20	25	10	12	6	10		5	4	3	10
ACTIVITY	1-2	1-3	2-3	2-4	3-4	5-4	5-3															
TIME	20	25	10	12	6	10																

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

Estd. 1980

AUTONOMOUS

NOTE : Questions can be given as A, B splits or as a single Question for 10 marks

IV B.Tech. I Semester MODEL QUESTION PAPER

IoT Applications

For CIC

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	What is IoT convergence?	1	2	2
	b).	What is meant by IoT standardization?	1	2	2
	c).	What is IoT value chain?	2	2	2
	d).	State the main design principles of IoT architecture.	2	2	2
	e).	Define functional view in IoT architecture.	3	2	2
	f).	What is state-of-the-art IoT architecture?	3	2	2
	g).	Define Brownfield IoT.	4	2	2
	h).	What is serialization in IoT?	4	2	2
	i).	What is trust management in IoT?	5	2	2
	j).	List security issues in IoT.	5	2	2

5 x 10 = 50 Marks

UNIT-1					
2.	a).	Describe the IoT vision and future Internet technologies.	1	2	5
	b).	Describe the importance of security and privacy in IoT systems.	1	2	5
OR					
3.	a).	Discuss IoT infrastructure and communication networks.	1	2	5
	b).	Explain IoT research directions and standardization activities.	1	2	5
UNIT-2					
4.	a).	Explain the structure of IoT value chains.	2	3	5
	b).	Explain the main design principles of IoT architecture.	2	3	5
OR					
5.	a).	Discuss the industrial structure for IoT systems.	2	3	5
	b).	Describe the role of standards in IoT architecture design.	2	3	5
UNIT-3					
6.	a).	Explain the IoT reference model with its components.	3	3	5
	b).	Describe the information view and deployment view in IoT.	3	3	5
OR					

7.	a).	Explain the functional view in IoT architecture	3	3	5
	b).	Discuss operational view and other architectural views in IoT systems.	3	3	5
		UNIT-4			
8.	a).	Examine IoT applications in smart factory systems.	4	4	5
	b).	Discuss smart objects and smart applications in IoT.	4	4	5
		OR			
9.	a).	Examine the role of IoT in oil and gas industry.	4	4	5
	b).	Examine IoT applications in home management and eHealth systems.	4	4	5
		UNIT-5			
10.	a).	Examine governance issues in IoT systems.	5	4	5
	b).	Discuss Smartie approach for secure IoT platforms.	5	4	5
		OR			
11.	a).	Examine security issues in IoT data platforms.	5	4	5
	b).	Examine data aggregation techniques used in smart city IoT systems.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



IV B.Tech. I Semester MODEL QUESTION PAPER

MALWARE ANALYSIS & REVERSE ENGINEERING

For CIC

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Explain the Malware Analysis Tools.	1	2	2
	b).	Discuss about resources of REM.	1	2	2
	c).	Demonstrate how to Identify Packers using PEiD	2	2	2
	d).	Illustrate PKI Certificates	2	2	2
	e).	List the Py commands	3	1	2
	f).	Explain the concept of Debugging	3	2	2
	g).	Discuss about code injection	4	2	2
	h).	Illustrate YARA to Identify Injected Code	4	2	2
	i).	Discuss How to Query Passive DNS	5	2	2
	j).	Demonstrate about the Interactive Maps.	5	2	2

Estd. 1980

AUTONOMOUS

5 x 10 = 50 Marks

		UNIT-I			
2.	a).	Explain the methodology of Reverse Engineering Malware (REM)	1	2	5
	b).	Describe Malware Classification and malware indicators with examples.	1	2	5
		OR			
3.	a).	Explain the importance of MA lab setup and key tools used	1	2	5
	b).	Describe the process of creating custom Clam AV databases.	1	2	5
		UNIT-II			
4.	a).	Explain the use of TSK for host and network discovery in malware forensics.	2	2	5
	b).	Apply Reg Ripper plugins for registry forensics with an example.	2	3	5
		OR			
5.	a).	Explain how PEiD is used to identify Packers.	2	2	5
	b).	Apply techniques to detect rogue PKI certificates and bypass file ACL restrictions.	2	3	5

		UNIT-III			
6.	a).	Explain the procedure of configuring JIT Debugger for shellcode analysis.	3	2	5
	b).	Apply debugging techniques with Python scripts for analyzing malware.	3	3	5
		OR			
7.	a).	Explain how to debug DLLs in a VMware/Parallels environment.	3	2	5
	b).	Apply breakpoint setting and execution control in malware debugging.	3	3	5
		UNIT-IV			
8.	a).	Explain the steps involved in memory dumping using MoonSols Windows Memory Toolkit	4	2	5
	b).	Discuss how Volatility helps in investigating processes from memory dumps.	4	2	5
		OR			
9.	a).	Describe the process of identifying injected code in memory using YARA.	4	2	5
	b).	Explain the methodology to detect and analyze suspicious DLLs loaded in memory.	4	2	5
		UNIT-V			
10.	a).	Explain DNS hostname resolution and passive DNS query techniques.	5	2	5
	b).	Apply WHOIS and Reverse IP Search for domain research and mapping.	5	3	5
		OR			
11.	a).	Explain how to create static and interactive IP maps for source tracking.	5	2	5
	b).	Apply DNS record analysis to trace malware source origins	5	3	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23C14104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
CRYPTO CURRENCY TECHNOLOGIES					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain About Digital Money	1	2	2
	b).	What is Digital Signature	1	2	2
	c).	Explain about Mining in Block Chain	2	2	2
	d).	What is Transaction Fee	2	2	2
	e).	Explain about Bitcoin Cash	3	2	2
	f).	Explain about Altcoins.	3	2	2
	g).	What is Consensus Algorithm	4	2	2
	h).	Explain about Smart Contracts.	4	2	2
	i).	Explain about crypto currency exchange	5	2	2
	j).	What is Domain Name Service	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.		Discuss about Interbank Payment Modes	1	2	10
OR					
3.		Explain about different types of Encryptions with examples.	1	2	10
UNIT-2					
4.		Explain blockchain advantage over conventional distributed database.	2	2	10
OR					
5.		Discuss about Distributed Consensus, Merkle Patricia Tree and Gas Limit.	2	2	10
UNIT-3					
6.		Explain the complete process related to working of Bitcoin	3	2	10
OR					
7.		Demonstrate the process of storing and usage of Bitcoin and Mining.	3	2	10
UNIT-4					
8.		Explain about Ethereum smart contracts using Solidity Language	4	2	10
OR					
9.		Explain applications of cryptocurrencies and blockchain.	4	2	10

UNIT-5					
10.		Explain about Crypto currency Regulations and Legal Aspects.	5	2	10
OR					
11.		Explain about Blockchain Applications of Internet of Things and Medical Record Management System.	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 mark



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Course Code: B23C14105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Design of IoT Architecture					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the role of IoT architecture.	1	2	2
	b).	State the purpose of time bases in IoT.	1	2	2
	c).	Mention the influence of power consumption in IoT systems.	2	2	2
	d).	Distinguish between single-hub and multi-hub networks.	2	2	2
	e).	Write the significance of Industry 4.0.	3	2	2
	f).	Mention any two applications of IIoT.	3	2	2
	g).	Mention generic application security in IoT.	4	2	2
	h).	Write reliable-and-secure-by-design IoT applications.	4	2	2
	i).	Mention the role of fuzzing in industrial control network systems.	5	2	2
	j).	State the purpose of Modbus/TCP fuzzer.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the IoT landscape with suitable applications.	1	2	5
	b).	Discuss protocol concepts and IoT-oriented protocols.	1	2	5
		OR			
3.	a).	Explain the role of databases and time bases in IoT systems.	1	2	5
	b).	Discuss the role of wireless networks in IoT communication.	1	2	5
		UNIT-2			
4.	a).	Interpret the IoT device design space with suitable factors.	2	3	5
	b).	Explain the relationship between chip size, duty cycle and device performance.	2	3	5
		OR			
5.	a).	Compare single-hub and multi-hub networks in IoT.	2	3	5
	b).	Explain environmental interaction modeling and event transport in IoT systems.	2	3	5
		UNIT-3			
6.	a).	Explain Industrial Internet of Things architecture with suitable context.	3	3	5
	b).	Discuss major applications of IIoT in real-time industries.	3	3	5

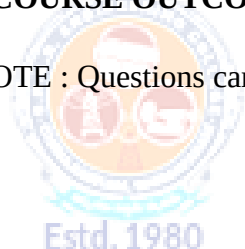
		OR			
7.	a).	Explain how IIoT supports smart manufacturing.	3	3	5
	b).	Discuss the challenges involved in implementing IIoT systems.	3	3	5
		UNIT-4			
8.	a).	Examine the security requirements of IoT systems.	4	4	5
	b).	Analyze and write the generic application security and process security in IoT.	4	4	5
		OR			
9.	a).	Examine the role of run-time monitoring in secure IoT systems.	4	4	5
	b).	Analyze and write the ARMET approach in IoT safety and security.	4	4	5
		UNIT-5			
10.	a).	Examine fuzz testing for IoT security.	5	4	5
	b).	Analyze and write the working of Modbus/TCP fuzzer.	5	4	5
		OR			
11.	a).	Examine the importance of fuzz testing in identifying vulnerabilities in IoT systems.	5	4	5
	b).	Analyze and write the fuzzing in industrial control network systems.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23CI4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
CYBER LAWS AND SECURITY POLICIES					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Computer Security.	1	1	2
	b).	List any two types of security threats.	1	1	2
	c).	What is Confidentiality in information security?	2	1	2
	d).	Explain the purpose of access control.	2	2	2
	e).	What is the Orange Book?	3	1	2
	f).	Define Security Policy.	3	1	2
	g).	Explain about asset classification?	4	2	2
	h).	Mention any two employee responsibilities in security.	4	1	2
	i).	Explain about Information Processing Security?	5	2	2
	j).	Define Human factors in security.	5	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain the fundamental concepts of computer security and its importance.	1	2	5
	b).	Describe different types of security threats and vulnerabilities.	1	2	5
		OR			
3.	a).	Explain government requirements and standards in computer security.	1	2	5
	b).	Explain access control mechanisms for information protection with examples.	1	2	5
		UNIT-II			
4.	a).	Apply secure system planning and administration issues for real world examples.	2	3	5
	b).	Demonstrate the role of Orange Book in system security evaluation.	2	3	5
		OR			
5.	a).	Apply security policy requirements to design a system that ensures accountability and assurance in an organization.	2	3	5
	b).	Apply network security concepts in real-world scenarios.	2	3	5
		UNIT-III			
6.	a).	Explain the different types of corporate security policies (Tier 1, Tier 2,	3	2	5

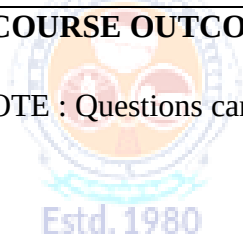
		Tier 3).			
	b).	Explain steps involved in developing security policies.	3	2	5
		OR			
7.	a).	Explain process management in information security policies.	3	2	5
	b).	Explain asset classification policy with suitable examples.	3	2	5
		UNIT-IV			
8.	a).	Explain fundamentals of information security and employee responsibilities.	4	2	5
	b).	Apply information classification and handling procedures.	4	3	5
		OR			
9.	a).	Explain tools used in information security.	4	2	5
	b).	Demonstrate secure program administration with examples.	4	3	5
		UNIT-V			
10.	a).	Explain the role of Information Security Management Standards.	5	2	5
	b).	Apply human factors in improving organizational security.	5	3	5
		OR			
11.	a).	Discuss the role of information security professionals.	5	2	5
	b).	Demonstrate adoption of security standards in organizations.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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IV B.Tech. I Semester MODEL QUESTION PAPER

INTRUSION DETECTION AND PREVENTION SYSTEM

For CIC

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define Intrusion Detection System (IDS).	1	1	2
	b).	What is the primary difference between IDS and IPS?	1	2	2
	c).	Explain about Hybrid IDS	2	2	2
	d).	Differentiate between credential analysis non credential analysis	2	2	2
	e).	What is Snort?	3	1	2
	f).	Discuss the Snort alert modes.	3	2	2
	g).	Explain about Rule headers	4	2	2
	h).	Explain how to configure snort file	4	2	2
	i).	Discuss about using ACID with Snort	5	2	2
	j).	Draw the architecture model of IDS.	5	2	2

5 x 10 = 50 Marks

		UNIT-I			
2.	a).	Explain the basic concepts of Intrusion Detection Systems (IDS) and describe the need and types of IDS.	1	2	5
	b).	Differentiate between host-based and network-based information sources in IDS.	1	2	5
		OR			
3.	a).	Define and explain various internal and external threats to data in an organization.	1	2	5
	b).	Discuss the history and evolution of intrusion detection systems.	1	2	5
		UNIT-II			
4.	a).	Demonstrate the working principles of Intrusion Prevention Systems (IPS) and Hybrid IDS.	2	3	5
	b).	Apply different response types and the process of mapping responses to policies in IDS.	2	3	5
		OR			
5.	a).	Implement credential and non-credential vulnerability analysis techniques.	2	3	5

	b).	Apply protocol-based IDS and explain its analysis scheme.	2	3	5
		UNIT-III			
6.	a).	Demonstrate the step-by-step procedure to compile and install Snort with multiple network interfaces.	3	3	5
	b).	Apply various Snort alert modes and explain with examples.	3	3	5
		OR			
7.	a).	Implement Snort command line options and scenarios for installing Snort in real-time environments.	3	3	5
	b).	Use different Snort modes and explain their functionalities.	3	3	5
		UNIT-IV			
8.	a).	Demonstrate Snort rule headers and options with suitable examples.	4	2	5
	b).	Apply the usage of plugins, preprocessors, and output modules in Snort.	4	3	5
		OR			
9.	a).	Implement how to configure Snort with MySQL and discuss its advantages.	4	3	5
	b).	Apply the structure and significance of Snort configuration files.	4	3	5
		UNIT-V			
10.	a).	Explain the role of ACID and Snort Snarf with Snort in intrusion detection.	5	2	5
	b).	Demonstrate the effectiveness of using Snort with ACID in real-time intrusion detection scenarios.	5	3	5
		OR			
11.	a).	Discuss the architecture models of IDS and IPS with appropriate diagrams.	5	2	5
	b).	Use agent development for intrusion detection and Describe the steps involved in it.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23CI4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
METAVERSE					
(Common to CIC, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate between VR and AR.	1	2	2
	b).	Apply the concept of Blockchain in virtual transactions	1	3	2
	c).	Explain the role of sensors in VR systems.	2	2	2
	d).	Apply VR technology in training simulations.	2	3	2
	e).	How are NFTs different from cryptocurrencies?	3	2	2
	f).	Name one blockchain platform.	3	3	2
	g).	Explain the concept of digital goods with examples	4	2	2
	h).	Explain how digital identity works in the Metaverse	4	2	2
	i).	State the purpose of WebXR in Metaverse applications	5	2	2
	j).	What is the purpose of interoperability standards?	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the applications of the Metaverse in gaming and entertainment.	1	3	5
	b).	Analyze the challenges faced in the development of the Metaverse.	1	4	5
OR					
3.	a).	Explain the role of the Metaverse in healthcare with examples.	1	3	5
	b).	Compare VR, AR, and MR technologies in terms of features and applications	1	4	5
UNIT-2					
4.	a).	Describe hardware and software requirements for VR/AR.	2	2	5
	b).	Analyze the role of Metaverse platforms like Oculus and Microsoft Mesh	2	4	5
OR					
5.	a).	Explain interaction techniques used in VR systems.	2	2	5
	b).	Evaluate challenges in designing VR/AR systems.	2	4	5
UNIT-3					
6.	a).	Explain how Blockchain technology works.	3	2	5
	b).	Analyze the role of decentralization in the Metaverse ecosystem	3	4	5

		OR			
7.	a).	Describe the working of Smart Contracts in blockchain systems	3	2	5
	b).	Analyze security and privacy issues in Blockchain systems.	3	4	5
		UNIT-4			
8.	a).	Explain the Play-to-Earn (P2E) model with suitable examples.	4	2	5
	b).	Discuss how Decentralized Finance (DeFi) supports the Metaverse economy.	4	3	5
		OR			
9.	a).	Explain the concept of tokenomics and its key components.	4	2	5
	b).	Analyze ethical issues such as privacy and security in the Metaverse.	4	4	5
		UNIT-5			
10.	a).	Explain the features and applications of Unity in Metaverse development.	5	2	5
	b).	Analyze the advantages and limitations of WebXR	5	4	5
		OR			
11.	a).	Explain the role of Artificial Intelligence in enhancing user experience in the Metaverse	5	2	5
	b).	Analyze how AI and cloud computing work together in Metaverse platforms.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CI4110					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SMART CONTRACT ESSENTIALS					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is Smart Contract?	1	2	2
	b).	Explain the concept of Pattern.	1	2	2
	c).	What is Ethereum Wallet?	2	2	2
	d).	Explain the usage of Ethereum Key pairs.	2	2	2
	e).	What is Distributed Ledger?	3	2	2
	f).	Explain about DAO	3	2	2
	g).	Write about Roots of Bitcoin	4	2	2
	h).	Explain the functionality of Crypto currency Exchange	4	2	2
	i).	What is Digital Signature?	5	2	2
	j).	Explain the Limitations of Bitcoin.	5	2	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the Structure of Smart Contract and Interaction with Examples.	1	2	10
		OR			
3.	a).	Explain the Patterns used in Smart Contracts with Examples	1	2	10
		UNIT-2			
4.	a).	Demonstrate the Ethereum Development Tools.	2	2	10
		OR			
5.	a).	Explain about Ethereum Accounts and Ethereum Key pairs.	2	2	10
		UNIT-3			
6.	a).	Demonstrate the Mining strategy and Rewards associated with Mining.	3	2	10
		OR			
7.	a).	Explain about different types of Vulnerabilities and Attacks.	3	2	10

		UNIT-4			
8.	a).	Demonstrate about the Roots of Bitcoin and Legal Aspects.	4	2	10
		OR			
9.	a).	Explain about how smart contracts can be used in Medical Record Management.	4	2	10
		UNIT-5			
10.	a).	Explain about Bitcoins Transactions and Bitcoins Network.	5	2	10
		OR			
11.	a).	Explain the process of Bitcoins Payments and APIs.	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23CI4111					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUGUMENTED REALITY & VIRTUAL REALITY					
For CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write brief notes on the history of augmented reality.	1	3	2
	b).	Explain the characteristics of tracking technology.	1	3	2
	c).	Explain Outdoor Tracking.	2	3	2
	d).	Explain Scene Graphs.	2	3	2
	e).	Explain the human physiology and perception in VR.	3	3	2
	f).	Explain the optical aberrations present in virtual reality (VR).	3	3	2
	g).	Discuss the implications for virtual reality (VR).	4	3	2
	h).	Explain how to correct optical distortions.	4	3	2
	i).	The Vestibular System: An Explanation.	5	3	2
	j).	Explain the Auditory Rendering.	5	3	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain augmented reality (AR), provide examples of AR applications, and discuss related fields such as virtual reality (VR) and mixed reality (MR).	1	3	5
	b).	Explain visual perception in augmented reality (AR) and define what multimodal displays are in AR.	1	3	5
OR					
3.	a).	Explain the requirements and characteristics of AR displays, as well as the spatial display model and visual displays.	1	3	5
	b).	Discuss the coordinate systems used in AR tracking, as well as the differences between stationary tracking systems and mobile sensors.	1	3	5
UNIT-2					
4.	a).	How does AR marker track work? Describe briefly the advantages of tracking natural features.	2	3	5
	b).	Describe the types of AR interaction modalities and the difficulties in outside AR tracking.	2	3	5
OR					
5.	a).	Describe the following interaction modalities: haptic interaction in augmented reality, virtual user interfaces on real surfaces and	2	3	5

		augmented paper, output, input, and tangible interfaces.			
	b).	Describe the function of scene graphs in augmented reality applications.	2	3	5
		UNIT-3			
6.	a).	Discuss the virtual reality (VR) and A synopsis of human physiology and perception in virtual reality.	3	3	5
	b).	Illustrate axis-angle rotations, geometric models, VR viewing transformations, and light and optics, including lenses, aberrations, the human eye, cameras, and displays.	3	3	5
		OR			
7.	a).	Explain virtual reality (VR) and its main features. How does human perception affect VR design?	3	3	5
	b).	Define the effects of lenses and optical aberrations. VR. How do cameras and displays function in VR systems?	3	3	5
		UNIT-4			
8.	a).	Identify the routes that lead from photoreceptors to the cortex of the eye. How do the cornea and photoreceptors of the human eye process light?	4	2	5
	b).	Demonstrate eye movements and how they affect VR design, depth, motion, and color perception.	4	3	5
		OR			
9.	a).	Explain the visual cortex's function in perception. How do rods and cones, which are photoreceptors, function? How does virtual reality use depth perception?	4	3	5
	b).	Explain visual rendering techniques such as ray tracing, shading, rasterization, correcting optical aberrations and latency in virtual reality, and creating immersive images and videos for VR.	4	3	5
		UNIT-5			
10.	a).	Outline the visual pathways that connect photoreceptors to the visual cortex. How does light travel from cornea to photoreceptors?	5	3	5
	b).	Explain eye movements, types, and consequences for VR design and in-depth perception of VR cues (binocular disparity, etc.).	5	3	5
		OR			
11.	a).	Describe how motion is perceived. How VR makes use of motion perception, color perception, the influence of color vision in VR, and rendering in VR: ray tracing versus rasterization.	5	3	5
	b).	Discuss the correction for distortion. How and why is it done in virtual reality? Outline latency reduction and methods for enhancing virtual reality.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks